

MACHINE LEARNING PROJECT REVIEW

Enterprise Knowledge Intelligence Platform

*A Hybrid Relational, Document, and Vector Data Architecture
for Governed, Semantic Enterprise Knowledge Retrieval*

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Motivation & Literature Review

Enterprise knowledge — SOPs, technical specs, institutional know-how — is scattered across file shares, wikis, and chat logs. Legacy single-database systems can't equally serve structured, semi-structured, and unstructured content, producing two recurring failures: poor discoverability (keyword search misses meaning) and weak governance (access control too permissive or too rigid).

1 Data Storage Paradigms

Kimball & Ross (2013)

Relational modeling — strong consistency; rigid schema limits fit for evolving content.

Sadalage & Fowler (2012)

Polyglot persistence — no single DB is optimal; combining stores adds sync overhead.

MongoDB (2023)

Schema-flexible document storage — weaker joins & transactional guarantees than SQL.

2 Semantic Search & Retrieval

Reimers & Gurevych (2019)

Sentence-BERT — embeddings where similarity maps to vector distance.

Malkov & Yashunin (2020)

HNSW graphs enable efficient ANN search over millions of vectors.

Lewis et al. (2020)

RAG — semantic retrieval feeding relevant context into language tasks.

3 Access Control & API Security

Ferraiolo et al. (2007)

RBAC balances expressiveness and administrative simplicity.

Hardt, IETF RFC 6749 (2012)

OAuth 2.0 — standardized delegated authorization for REST APIs.

Jones et al., RFC 7519 (2015)

JWTs — compact, self-contained tokens for stateless sessions.

Critical Analysis & Research Gaps

Each research strand solves one dimension well but leaves the others unaddressed in isolation — no reviewed work integrates storage flexibility, semantic retrieval, and governed access together. That gap defines the space EKIP occupies.

Storage Literature

✓ Solves structure vs. flexibility

✗ Silent on retrieval quality or security

Retrieval Literature

✓ Solves lexical vs. semantic matching

✗ Assumes an open, ungoverned corpus

Security Literature

✓ Solves authorization & stateless auth

✗ Assumes content is already organized

G1

No unified governance

Access control enforced per-database, not consistently across relational, document, and vector data.

G2

Retrieval assumes open corpus

Semantic search research rarely resolves results against per-user permissions before returning content.

G3

Weak canonical linkage

Polyglot-persistence work lacks a concrete strategy for keeping three data stores referentially consistent.

G4

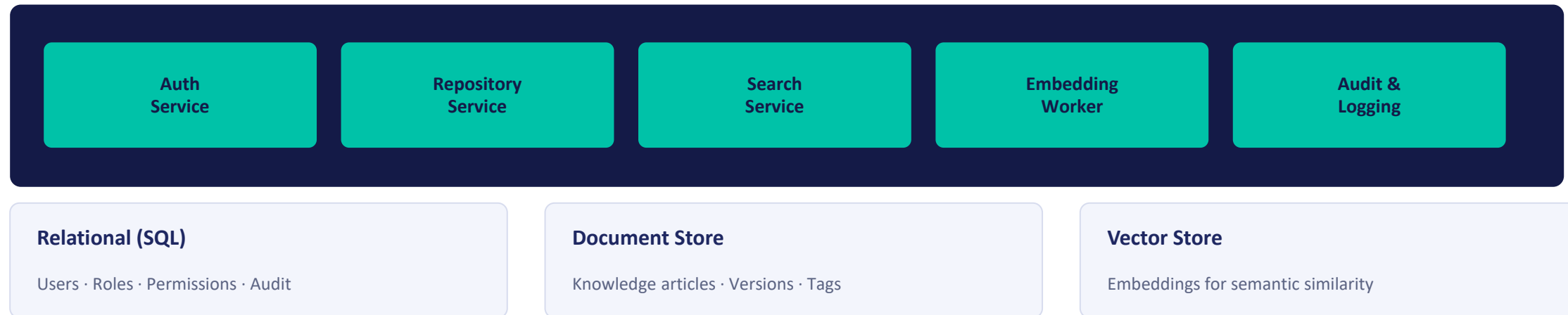
Limited hybrid evaluation

Few studies compare keyword-only, semantic-only, and hybrid re-ranked search within one governed system.

Proposed Architecture & Method

EKIP unifies relational, document, and vector storage behind one governed API. A canonical document UUID links every record across all three stores; every search result is resolved against the permissions table before it is returned.

Client Apps → API Gateway (Auth · Routing · Rate Limiting)



Canonical document UUID links every record across all three stores.

Semantic Search Request Flow



Evaluation & Conclusion

Precision@k / Recall@k

Hybrid search benchmarked against keyword-only and semantic-only modes

API Latency

Response time measured under simulated concurrent load

Search Throughput

Queries served per second as replicas scale horizontally

100% Access Compliance

Every result resolved through the permissions table before return

EKIP unifies relational, document, and vector storage behind a single, secure REST API — closing the gap between governed access and meaning-aware retrieval that isolated prior work leaves open.

Future Directions

- RAG for automatic Q&A over the knowledge base
- Feedback-driven re-ranking to improve relevance
- Connectors to email, chat, and ticketing systems
- Full empirical evaluation on production-scale data